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SUPPLEMENTARY MATERIAL

The Sierpiński Blocks of G's Graded Operator Are a Null Rotation

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Abstract

Gravity Field Dynamics writes Newton's constant as $G = \frac{17}{13}\,\alpha^{21}\,\hbar c/m_e^2$, read from a graded operator M_s on blocks of size 1, 4 and 16 whose traces are the coupling polynomial $f(K,s) = 1 + sK + K^2$. The archive draws the blocks as Sierpiński triangles, $S_m = A^{\otimes m}$ with $A = \left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$, on $m = 0, 2, 4$ bits. G reads only their diagonal. Reading the two bits of the 4-block as spinor indices, A acts on 4-vectors as a null rotation, and we show exactly what M_s is at each state: at $s = +1$ (TetA) it is the null rotation acting on scalar, vector and rank-2 tensor; at $s = -1$ (TetB) it is that rotation composed with the total inversion -1, not with spatial parity; at $s = 0$ (the Field Origin) it is their average, the projection onto the part even under -1. The graded trace is the character $1 + s\chi + \chi^2$ with $\chi = |\operatorname{Tr} A|^2 = 4 = K$; among Lorentz transformations whose trace is real, $\chi = 4$ only for ± 1 and the null rotations. The Walsh–Hadamard read on the 4-block is a rotation of space by π and factors as null rotation, boost of rapidity $\ln 2$ with a π twist, and reverse null rotation. The Jordan chains split 16 into $(5+3+1+1)+(3+3)$, the pieces of a metric and of a 2-form, and the archive's tetrad frame, read in the same way, is a Newman–Penrose null tetrad. The results identify the operator G is read from; they do not derive G or K.

Keywords: gravitational constant, Sierpiński triangle, Walsh–Hadamard transform, null rotation, Lorentz group, coupling polynomial, proof

Setting

The archive derives G from the graded operator $I_s = P_0 + sP_1 + P_2$ on blocks of size 1, $K = 4$ and $K^2 = 16$, the scalar, vector and tensor parts of the tetrad equation [1]. Its traces are $f(K,s) = 21, 17, 13$ at $s = +1, 0, -1$ (TetA, the Field Origin, TetB), and $\det(\alpha^{I_s}) = \alpha^{\operatorname{Tr} I_s}$ gives $G = \frac{17}{13}\,\alpha^{21}\,\hbar c/m_e^2$. The archive's diagram of this operator [2] draws the three blocks as Sierpiński triangles, and its account of the cube says that the blocks "are" the Sylvester–Hadamard matrices H_1, H_4, H_{16} [3]. We take the diagram's blocks at face value and determine what they are.

Contributions and prior work. The spinor–vector correspondence, null rotations [5], the classification of $SL(2,\mathbb{C})$ by its trace [9], the Jordan structure of a nilpotent element of $\mathfrak{sl}_2$ [8], null tetrads built from a spin dyad [6], Lucas's theorem [7] and the triangular factorisation of the Walsh–Hadamard matrix are all standard, and we claim no novelty for them. What is new is the identification of these structures with the blocks of G's operator and with the archive's tetrad frame, and the exact form of M_s at the three states (Theorem 5). In particular, the TetB sign acts on the vector block as the total inversion -1, not as spatial parity. Write $A = \left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$, $H_2 = \left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$, $S_m = A^{\otimes m}$, and let $|x|$ be the number of 1-bits of an index x.

The Blocks

Theorem 1 (the Sierpiński blocks). *The diagram's blocks are S_0, S_2 and S_4 (checked cell by cell; the blocks are defined by $S_m = A^{\otimes m}$). Entry (i,j) of S_m is 1 exactly when the bits of j are a subset of those of i, that is, when $\binom{i}{j}$ is odd. With $L = \sum_b N_b$, where N_b removes bit b (it acts as $\left(\begin{smallmatrix}0 & 0\\ 1 & 0\end{smallmatrix}\right)$ on that bit),*

$$S_m = e^L, \qquad L^{m+1} = 0, \qquad \operatorname{Tr} L = 0. \tag{1}$$

The traces are $1, 4, 16$ and the filled areas are $1, 9, 81 = 3^m$.

Proof. The diagram's cells are compared with S_m one by one. $A = e^N$ with $N^2 = 0$, and the N_b commute, so $A^{\otimes m} = \prod_b e^{N_b} = e^L$. The entry rule is the tensor product of A's and Lucas's theorem. Each bit contributes 1 to the trace and 3 to the area. □

Theorem 2 (G reads the diagonal). *Let $M_s = S_0 \oplus s\,S_2 \oplus S_4$. Then $\operatorname{Tr} M_s = 1 + 4s + 16 = f(K,s)$, the spectra of M_s and I_s coincide (rank 17 at $s = 0$; 17 eigenvalues $+1$ and 4 eigenvalues -1 at $s = -1$), and $\det(\alpha^{M_s}) = \alpha^{f(K,s)}$. So G is the same whether the blocks are identities or Sierpiński blocks.*

Proof. Each S_m is lower triangular with unit diagonal, so its eigenvalues are all 1. $\det e^X = e^{\operatorname{Tr} X}$. □

So the Sierpiński structure off the diagonal is invisible to G. For comparison only: reading the filled cells instead, $1 + 9 + 81 = 91$, would change G by $\alpha^{70} \approx 10^{-150}$. The next theorems say what the off-diagonal structure is.

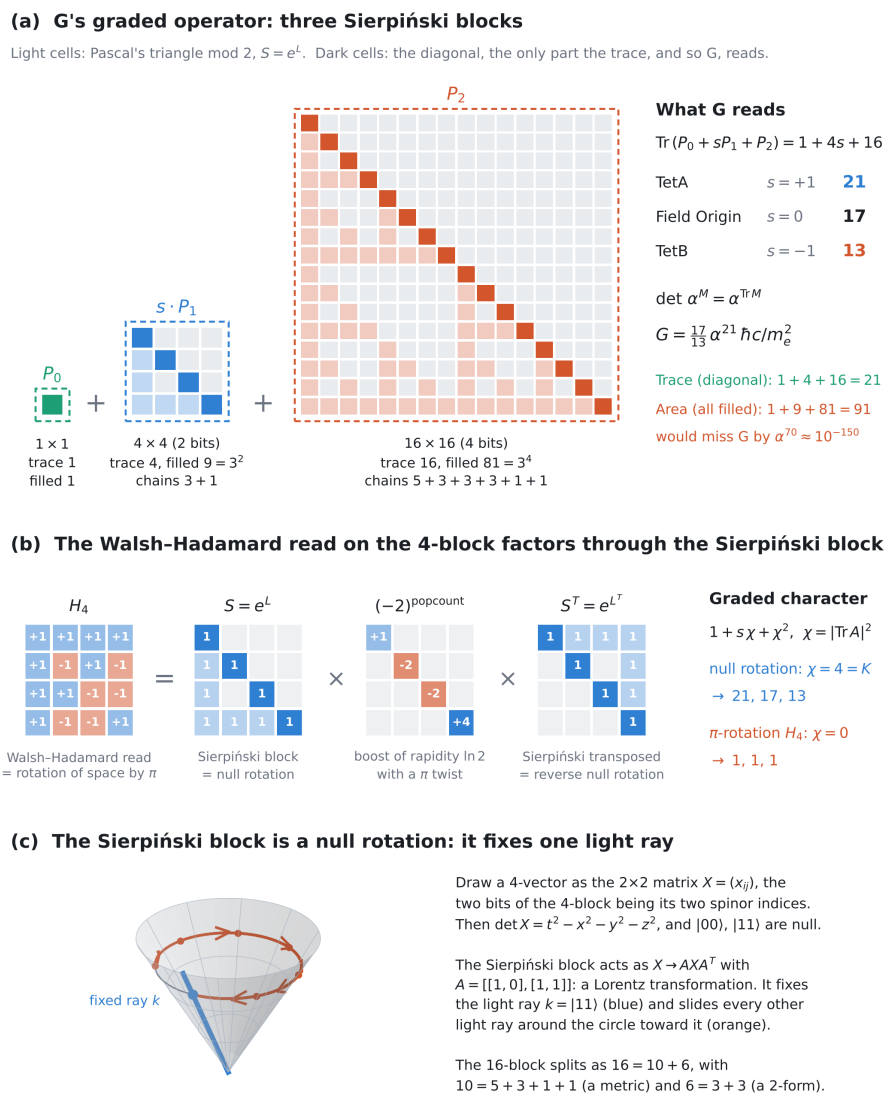


Figure 1. (a) G's graded operator as drawn in the archive: the Sierpiński blocks S_0, S_2, S_4. Their diagonals (dark) are all that the trace, and so G, reads. (b) The Walsh–Hadamard read on the 4-block factors through the Sierpiński block (Theorem 4). (c) Read as acting on 4-vectors, the Sierpiński block fixes the light ray $|11\rangle$ (Theorem 3); the one-parameter group of null rotations $A(\tau)$ through it carries every other light ray to it as $\tau \to \infty$, shown on the slice $y = 0$, which it preserves.

The Lorentz Reading

Write a 4-vector as $X = t + x\sigma_1 + y\sigma_2 + z\sigma_3$, so that $\det X = t^2 - x^2 - y^2 - z^2$, and index its entries by the two bits of the 4-block, $X = (x_{ij})$ [5]. The states $|00\rangle$ and $|11\rangle$ are $t + z$ and $t - z$, the two null directions along z.

Theorem 3 (the Sierpiński block is a null rotation). *On the bit basis,* $S_2 \operatorname{vec}(X) = \operatorname{vec}(AXA^T)$. *The map* $\Lambda_A : X \mapsto AXA^\dagger$ *is a proper orthochronous Lorentz transformation with* $\Lambda_A \neq 1$ *and* $(\Lambda_A - 1)^3 = 0$*: a null rotation. Its fixed vectors form the plane spanned by* $k = (1, 0, 0, -1)$*, the state* $|11\rangle$*, and the* y *axis, and* k *is the only light ray in that plane. Every real* A *with* $\det A = 1$ *fixes* y*, so the Sierpiński blocks act as the Lorentz group of* (t, x, z).

Proof. $\operatorname{vec}(AXA^T) = (A \otimes A)\operatorname{vec}(X)$, and A is real, so $A^T = A^\dagger$. $\det(AXA^\dagger) = \det X$ because $\det A = 1$. A fixes $|1\rangle$, so Λ_A fixes the matrix $|1\rangle\langle 1|$, which is the light ray k. The antisymmetric part $y\, i\sigma_2$ is sent to $\det A \cdot y\, i\sigma_2$. The remaining statements are computed from the 4×4 matrix of Λ_A. □

Remark (the one-parameter group). $A(\tau) = \left(\begin{smallmatrix} 1 & 0 \\ \tau & 1 \end{smallmatrix}\right)$ is the one-parameter group of null rotations about k. A boost along z conjugates it to itself with a new parameter, $B(\eta)A(\tau)B(\eta)^{-1} = A(\tau e^{-\eta})$, so every $A(\tau)$ with

$\tau > 0$ is equivalent to the Sierpiński block $A(1)$, and the entry 1 is a convention. As $\tau \to \infty$, $A(\tau)$ carries every other light ray to k.

Theorem 4 (the Hadamard read). *The normalised $H_2/\sqrt{2}$ acts on 4-vectors as a rotation of space by π about $(\hat{x}+\hat{z})/\sqrt{2}$. For $m = 0, 2, 4$,*

$$\begin{aligned} H_{2^m} &= S_m \operatorname{diag}\big((-2)^{|x|}\big)\, S_m^T \\ &= e^{L}\, e^{(\ln 2 + i\pi)|x|}\, e^{L^T}. \end{aligned} \tag{2}$$

On the 4-block the middle factor $\operatorname{diag}(1,-2)/\sqrt{2}$ is a boost of rapidity $\ln 2$ along z combined with a rotation by π about z. So the rotation by π is a null rotation, a boost with a π twist, and the reverse null rotation.

Proof. For one bit, $A \operatorname{diag}(1,-2)\, A^T = H_2$; tensor powers give the rest, and $(-2)^{|x|} = e^{(\ln 2 + i\pi)|x|}$. The Lorentz matrices follow from $\Lambda_M = \frac{1}{2}\mathrm{Tr}(\sigma_\mu M \sigma_\nu M^\dagger)$, with $\cosh \ln 2 = \frac{5}{4}$ and $\sinh \ln 2 = \frac{3}{4}$. □

This is an elementary triangular factorisation of the Walsh–Hadamard matrix (the matrix itself is Sylvester's [10]). In it, the Sierpiński blocks are the outer factors and the half-turn $e^{i\pi}$ sits in the middle.

The Character

Theorem 5 (the three states, and $K = 4$). *Let $\rho(V) = 1 \oplus V \oplus (V \otimes V)$ be the action of a Lorentz transformation V on scalar, vector and rank-2 tensor, and $\Lambda = A \otimes A$ the Sierpiński block in the bit basis. Then*

$$\begin{aligned} M_{+1} &= \rho(\Lambda), \qquad M_{-1} = \rho(-\Lambda), \\ M_0 &= \tfrac{1}{2}\big(M_{+1} + M_{-1}\big) = \rho(\Lambda)\,\Pi_+, \end{aligned} \tag{3}$$

where -1 is the total inversion (PT), which acts as -1 on vectors and $+1$ on tensors, and $\Pi_+ = \frac{1}{2}(1+\rho(-1))$ is the projection onto the part even under it. Spatial parity $P = \operatorname{diag}(1,-1,-1,-1)$ does not give TetB: the graded trace of $P\Lambda$ is 1. For $M \in SL(2,\mathbb{C})$ the trace of Λ_M on vectors is $\chi = |\operatorname{Tr} M|^2$ and on tensors χ^2, so $\operatorname{Tr}\rho(\pm\Lambda_M) = 1 \pm \chi + \chi^2 = f(\chi, \pm 1)$. The null rotation has $\chi = 4 = K$, giving $21, 17, 13$; the π-rotation $H_2/\sqrt{2}$ has $\chi = 0$. If $\operatorname{Tr} M$ is real, then $\chi = 4$ exactly when $M = \pm 1$ or M is a null rotation (up to sign); $\chi < 4$ for elements conjugate to rotations and $\chi > 4$ for those conjugate to boosts. Elements with non-real trace, which combine a boost and a rotation, can also have $\chi = 4$.

Proof. The blocks of $M_{\pm 1}$ are 1, $\pm(A \otimes A)$ and $(A \otimes A)^{\otimes 2} = S_4$, and $(-1)^{\otimes 2} = +1$ on tensors. Λ_M is $M \otimes \bar{M}$ in the bit basis, so its trace is $\operatorname{Tr} M \operatorname{Tr} \overline{M}$, and traces multiply over tensor products. $\operatorname{Tr} A = 2$, $\operatorname{Tr} H_2 = 0$, and $\operatorname{Tr}(P\Lambda_A) = -1$. The classification by trace is standard [9]: real trace in $(-2, 2)$ is elliptic, ± 2 is parabolic or ± 1, and real $|\mathrm{Tr}| > 2$ is hyperbolic. The loxodromic example $\operatorname{Tr} M = 2e^{i\pi/3}$ is in the checks. □

This gives G's three traces an exact reading: TetA is the null rotation on scalar, vector and tensor; TetB adds the total inversion; the Field Origin is the even projection, which is why $17 = (21+13)/2$. It does not derive $K = 4$, which comes from $D = 3$ [4], and the identity has the same character. It shows that the blocks the archive draws are null rotations, and that the Walsh–Hadamard read itself, with $\chi = 0$, could not give G's numbers.

Theorem 6 (the chains). *The Jordan chains of S_2 have lengths $3+1$, and those of S_4 have lengths $5+3+3+3+1+1$. They are the Clebsch–Gordan decompositions of 2 and 4 spin-$\frac{1}{2}$ factors, and over the whole operator $21 = 1 + (3+1) + (5+3+3+3+1+1)$. S_4 commutes with the exchange of the two vector factors, and on the symmetric 10 and antisymmetric 6 its chains are*

$$10 = 5+3+1+1, \qquad 6 = 3+3. \tag{4}$$

Proof. L is the lowering operator of the diagonal $\mathfrak{sl}_2$ on m spin-$\frac{1}{2}$ factors, so each irreducible summand of dimension d is one chain of length d. The multiplicity of dimension $m-2k+1$ is $\binom{m}{k} - \binom{m}{k-1}$. The restriction to the symmetric and antisymmetric parts is computed from the ranks of $(S_4 - 1)^j$. □

The split $16 = 10 + 6$ is the archive's: general relativity keeps the symmetric 10, and the antisymmetric 6 is the torsion-type part. The chains refine it into the pieces of a symmetric tensor $(5+3+1+1)$ and of a 2-form $(3+3)$.

The Tetrad Frame

Theorem 7 (the frame is the 4-block read). *The archive's tetrad frame [4], a row of ones over the four TetA corners $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$, is $H_2 \otimes H_2$ with its second and third rows exchanged. So it factors through the Sierpiński block,*

$$e = P\, S_2 \operatorname{diag}\big((-2)^{|x|}\big)\, S_2^T, \tag{5}$$

with P the row exchange: a null rotation, a boost of rapidity $\ln 2$ with a π twist, and the reverse null rotation. Read in (t,x,y,z), the frame is not a Lorentz structure: $e^T \eta\, e = 2J - 4I$, which is J minus the archive's forced Gram matrix $4I - J$.

Proof. Direct comparison, and Theorem 4 with $m = 2$. □

So the same 4×4 matrix plays two roles. As an operator on the 4-block it is a Lorentz transformation, the rotation by π. As a set of four legs it is the frame. The next theorem reads the legs in the spinor picture.

Theorem 8 (a Newman–Penrose null tetrad). *Read with its two bits as spinor indices, leg (a, b) of $H_2 \otimes H_2$ is the matrix $X = h_a h_b^T$, with h_0, h_1 the columns of H_2, so every leg has $\det X = 0$. In Minkowski components*

$$\begin{aligned} l &= (1, 1, 0, 0), \qquad n = (1, -1, 0, 0), \\ m &= (0, 0, -i, 1), \quad \bar{m} = (0, 0, i, 1), \end{aligned} \tag{6}$$

with $l \cdot n = 2$, $m \cdot \bar{m} = -2$ and every other product 0: a Newman–Penrose null tetrad [6], built from the spinor dyad (h_0, h_1), whose normalisation $\varepsilon(h_0, h_1) = -2$ accounts for the factor 2. Its Hadamard combinations $l \pm n$, $m + \bar{m}$ and $i(m - \bar{m})$ are $2t$, $2x$, $2z$ and $2y$. In any spinor reading, $X \mapsto BXC^T$ with B, C invertible, at most two of the four legs are real light rays.

Proof. The column (a, b) of $H_2 \otimes H_2$ is $h_a \otimes h_b = \mathrm{vec}(h_a h_b^T)$, a matrix of rank one. The components and products follow from $x^\mu = \frac{1}{2}\mathrm{Tr}(\sigma_\mu X)$ and the polarisation of det. For the bound: a rank-one matrix pq^T is Hermitian exactly when q is a real multiple of $\bar{p}$. If the legs $(a, 0)$ and $(a, 1)$ were both Hermitian, then Ch_0 and Ch_1 would both be multiples of $\overline{Bh_a}$, which is impossible for invertible C. So each a contributes at most one real light ray. □

So the frame's four legs are null in the spinor picture, in the form general relativity uses for null tetrads: two real light rays and one complex-conjugate pair. This is a different reading from Letter 1's. There the legs are $(1, v/\sqrt{3})$, the Hadamard frame with its spatial rows rescaled by $1/\sqrt{3}$, and all four are real light rays on the celestial sphere. Both readings make the frame null; they differ in which coordinates carry the two bits.

Scope

The results are exact. They identify the operator from which G is read; they do not derive G. The operator is a null rotation only at TetA; TetB composes it with the total inversion -1, and the Field Origin is a projection, not a Lorentz transformation. That the TetB sign acts on the vector block as PT, rather than as the spatial parity that exchanges TetA and TetB in three dimensions, is an open point for the framework's reading of the two chambers. The value of G still rests on the archive's assumption that each of the 21 coupling levels contributes one factor of α [1], and on $K = 4$ from $D = 3$. The trace cannot tell a null rotation from the identity, so the content added here is what the blocks are: Sierpiński triangles, null rotations, and the outer factors of the Walsh–Hadamard read. The statement that the blocks "are" H_1, H_4, H_{16} is sharpened accordingly: they are the Sierpiński factors of those matrices, and the matrices themselves have trace 0 beyond H_1. The Sierpiński blocks act through the Lorentz group of (t, x, z), with y fixed. So the chain lengths match the decomposition of a vector, a metric and a 2-form exactly, but naming the components (h_{00}, h_{0i}, …) requires the rotation group, which the Sierpiński block does not supply. The spinor–vector correspondence, null rotations and the factorisation of the Hadamard matrix are standard, as is the construction of a null tetrad from a spinor dyad; the identification with G's blocks and with the archive's tetrad frame is new. Theorem 8 gives two real null legs, not four, in the spinor reading. The Hadamard frame read directly in (t, x, y, z) has legs of norm -2; rescaling its spatial rows by $1/\sqrt{3}$, as Letter 1 does, makes all four legs real light rays.

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